FA 17.6: A Fully-Integrated Continuous-Time Programmable CCIR 601 Video Filter

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Digitizing video signals requires high-order anti-aliasing filters that handle large signal swings with low distortion. CCIR601 standards recommend the equi-ripple gain and group delay characteristics for filtering NTSC and PAL signals [1]. Currently, these characteristics are approximated by expensive hybrids that are not programmable for various applications. This paper introduces a method for integrating such filters using current-mode transimpedance-based structures. A single-chip solution demonstrated here matches the requirements with less than ±0.25dB gain ripple, ±20ns group delay variation in the passband, and more than 40dB attenuation in the stop band. The cut-off frequency, nominally set at 5.5MHz, is continuously programmable over a decade. The IC can drive 2Vpp signals into 75Ω load drawing 35mA quiescent current. Comparable integrated filters using other techniques require substantially larger currents [2].

The architecture of the complete filter as illustrated by Figure 1 is a 5th-order elliptic transfer function in tandem with a 3rd-order all-pass phase equalizer. The Cauer-elliptic response has an equiripple passband with a sharp roll-off into stop-band in the magnitude transfer function but causes excessive group delay peaking. The equalizer maintains this magnitude response while compensating for the group delay peaking. These two filters are represented in Figure 1 by a series of 2nd-order expressions that can be realized as biquads using transimpedance-based integrators.

Figure 2 represents the circuit diagram of the differential current mode amplifier [3] and a fully-differential opamp used for implementing the transimpedance integrator. The active cell draws current inputs into low impedance virtual ground nodes and conveys them to the intermediate nodes with a factor βR. These currents flow through the feedback capacitors around the high-input impedance differential amplifier establishing an integrated differential voltage output. The integrator time constant and, in turn, the filter frequency ω are set by βR, R and C as derived in Figure 2. There is no fundamental limit on the input voltage range as long as the bias currents are sufficient to provide for the resultant input current excursions. Since the input impedance remains constant over the full signal swings (Equation 2.1 in Figure 2) large-signal distortion is avoided and a wide dynamic range is achieved. This makes it possible to have large voltage swings. Transconductance-based Gm-C structures require large bias currents, in comparison, for low distortion [4]. The differential input/output amplifier requires appropriate common-mode feedback circuitry. Unity-gain emitter follower output stages provide low output impedance.

The basic cell is programmable in several ways. The differential input currents are conveyed to the pre-distortion diodes D1 and D2 that drive transistors N4 and N5 creating a differential output current Io±. The current gain βR of this Gilbert cell multiplier is set by the ratio of two DC currents (If/Is) that is accurately maintained over the temperature and process spreads. Scaling βR directly scales the filter frequency response. Output drive capability can be increased as necessary by increasing bias current Ib in the output drivers. Excess phase shift correction is implemented by a small resistance r_z in series with the feedback integrating capacitor that causes a phase lead. The excess phase shift correction achieved holds over a decade of frequencies, making the filters tunable over this range. In a complex filter, all cells may use the same input and output stages, feedback capacitors and bias currents. This achieves a uniform excess phase shift correction for all integrators. The excess phase shift can also be trimmed globally by varying the current Icomp. Various filter coefficients are obtained by simply ratioing the input and feedback resistors.

Using two integrators in feedback and feed-forward configuration, a generalized biquad with complex zeros is implemented as shown in Figure 3. The fully-differential configuration increases the dynamic range and reduces even-harmonic distortion. The parasitic capacitors are at low-impedance nodes. A realization of Figure 1 using minimum number of integrators is illustrated in Figure 4. The two biquadratic sections of the elliptic function are first cascaded and the real pole of the elliptic is then combined with the equalizer function to yield the same overall characteristics.

Elliptic poles and zeros give a flat magnitude response in passband and a 40dB roll-off from 5.5MHz to 8MHz. The equalizer transfer function corrects group delay to ±15ns to 90% of the cut-off frequency. These pole-zero values determine biquad coefficients and thus the resistor values in the biquads. Monte-Carlo simulations of the circuit show that even a 2% mismatch in the components does not degrade the filter performance, thus ensuring a high manufacturing yield. A supply-independent band-gap cell generates and distributes bias currents for all the transimpedance integrators. The cut-off frequency is programmed by globally scaling the current If, using a single external setting.

The entire filter, including the programmable bias generators, is integrated on an 8mm² chip using a 2μm complementary bipolar technology. The npn and pnp transistors have a cut-off frequency of 4GHz and 1.5GHz respectively. Gate-oxide-based capacitors and thin film resistors with 0.5% match set filter time constants. At 5.5MHz cut-off, the filter averages 2.5mA/pole. Nearly 15mA of the supply current is used for the output driver. The filter works from 5V rail-to-rail supplies to 10V. The measured frequency response of the single-ended filter output is shown in Figure 5. As demonstrated by Figure 6, the cut-off frequency is actually programmable from 250kHz-12.5MHz, with an external voltage control. Measured differential gain of 0.25% and differential phase of 0.2° are important for video applications.

Acknowledgments

The IC was fabricated at Raytheon Semiconductor Div., Mountain View, CA. Thanks to D. Ritter, J. Chu, D.J. Lee, C. Ly, V. Bahadhur, S. Zee and D. Shah.

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[2] Willingham, S. D., et al., "A BiCMOS Low-Distortion 8-MHz Low-Pass filter," IEEE J. Solid State Circuits, vol. 28, pp.1234-1245, Dec., 1993.

[3] Toumazon C., C. F. Ludgy, "Novel Bipolar Differential Input/Output Current Controlled Current Source," Electron Letters 21, pp. 199-200, 1985.

[4] Tsividis, Y.P., J. O. Voorman, editors, "Integrated Continuous Time filters: Principles, Design and Applications," IEEE Circuits and Systems Society selected reprint , IEEE Press, 1993.

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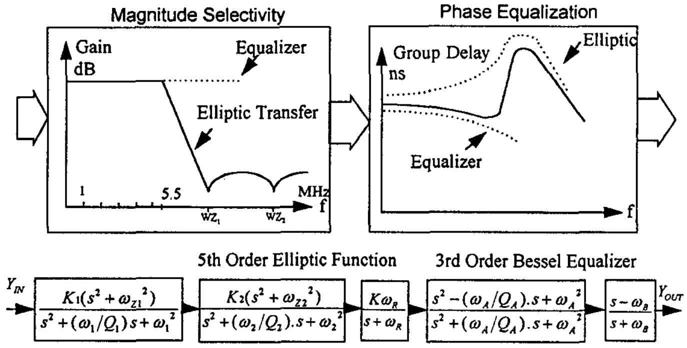
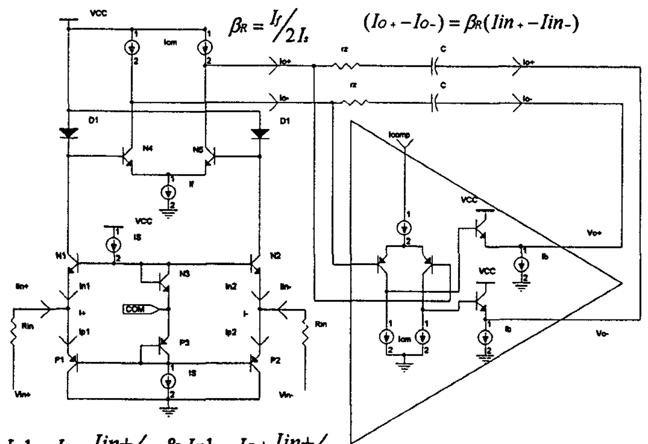


Figure 1: Composite video filter.



$$\Delta Iin = \Delta Vin / R$$

$$\Delta Io = \beta_R \Delta I_{IN}$$

$$\frac{\Delta Vo}{\Delta Vin} = \frac{(\beta_R/RC)}{s}$$

$$\omega = \frac{\beta_R}{RC}$$

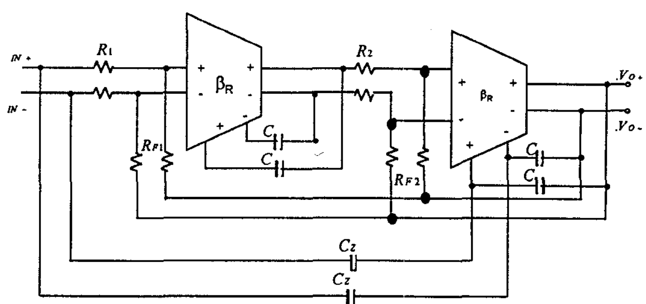
$$In1 = Is - {}^{Iin+}\!/_2 \quad \& \; Ip1 = Is + {}^{Iin+}\!/_2$$

$$gin = {}^1\!/_{rin} = {}^{In1}\!/_{V_T} + {}^{Ip1}\!/_{V_T} = {}^{2Is}\!/_{V_T} \quad = \text{Input Impedance} \quad ..\text{Eq.2.1}$$

Figure 2: Transimpedance integrator.

Supply voltage /current	**5V / 35mA**
Die area/feature size	**8mm²/2μm**
Maximum signal differential/single	**4Vpp/2Vpp**
Distortion @ 2Vpp	**<1%**
Diff gain/diff phase	**0.2dB/0.2°**
Passband ripple	**±0.25dB**
Group delay variation to 90% cut-off	**<±20ns**
Cut-off frequency accuracy	**5.5MHz±5%**
Cut-off frequency programmability	**<1M->10MHz**
Stop-band attenuation	**>40dB**

Table 1: Filter performance summary.



$$A_{DC} = R_{F1}/R_1 \qquad \omega_Z = \beta_R/\sqrt{CC_{Z1}R_1R_2} \qquad \frac{\Delta Vo}{\Delta V_{IN}} = A_{DC}\frac{(s^2/\omega_Z^2+1)}{(s^2/\omega_X^2 + s/Q\cdot\omega_X + 1)}$$

$$Q = \sqrt{\frac{R_{F2}}{R_{F1}}\cdot\frac{R_{F2}}{R_2}} \qquad \omega_X = \beta_R/\sqrt{R_{F1}R_2}\,C$$

Figure 3: Generalized pole-zero biquad.

Figures 4, 5, and 6: See page 383.

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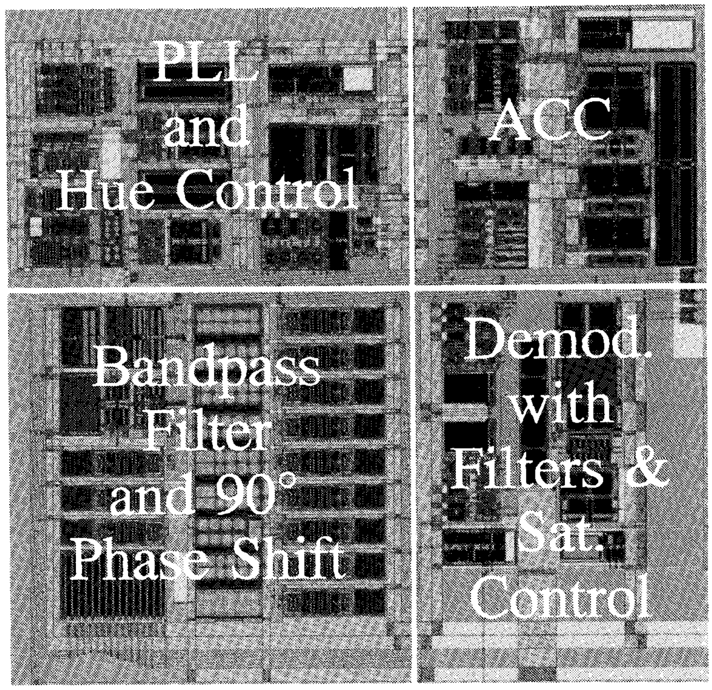

Figure 6: Chip micrograph of analog CMOS color-difference decoder.

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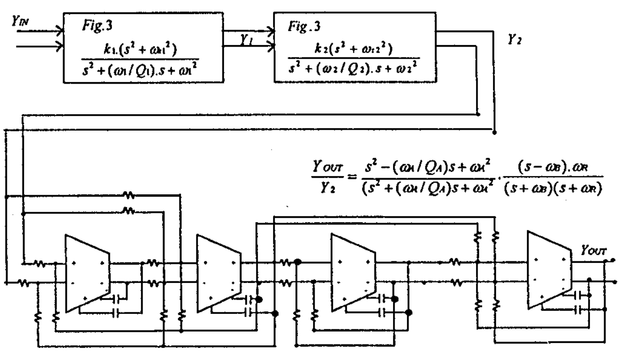

Figure 4: Complete video filter implementation .

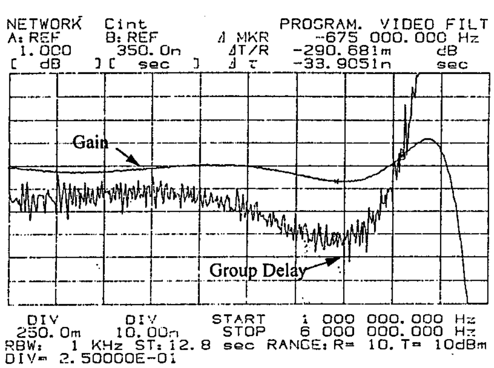

Figure 5: Measured gain and group delay response.

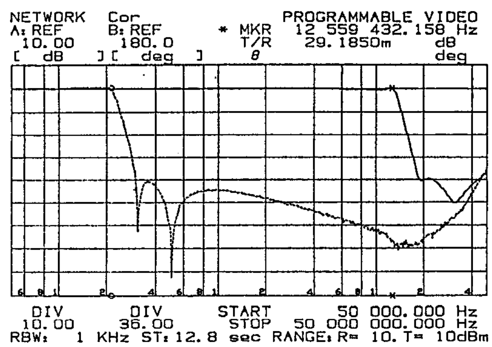

Figure 6: Measured frequency programming range.